

Law and Ethics in Information Security

Course Name : Information Security

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What is Information Security?

information Security is the protection of:

- Confidentiality
- Integrity
- Availability

Why Law & Ethics Matter:

- Protect users and organizations
- Prevent cybercrime
- Ensure responsible technology use
- Protect privacy and intellectual property

Legal and Ethical Aspects Overview

Four Major Areas:

1. Cybercrime and Computer Crime
2. Intellectual Property
3. Privacy
4. Ethical Issues

SECTION 1

Cybercrime and Computer Crime

What is Cybercrime?

Cybercrime refers to criminal activities involving computers or networks.

Computers can be:

As Targets

- Hacking
- Data theft
- Malware attacks

As Storage Devices

- Storing stolen passwords
- Pirated software
- Illegal files

As Communication Tools

- Online fraud
- Phishing
- Illegal online trading

Major Types of Cybercrime

- Illegal Access
- Data Interference
- System Interference
- Computer Fraud
- Identity Theft
- Phishing
- Malware Attacks
- Copyright Violations

Law Enforcement Challenges

- Lack of technical expertise
- Limited resources
- Cross-border crimes
- Difficult identification of criminals
- Low reporting rates by victims

Cybercriminals



Victims



Law Enforcement



Back to Cybercriminals

Working with Law Enforcement

Organizations should:

- Report incidents promptly
- Preserve evidence
- Cooperate with investigators
- Maintain security logs
- Follow legal procedures

SECTION 2

Intellectual Property

Intellectual Property (IP):

Human-created ideas and knowledge that have legal protection.

Examples:

- Software
- Databases
- Digital Content
- Algorithms

Types of Intellectual Property

Type	Protects
Copyright	Books, Software, Music
Patent	Inventions & Innovations
Trademark	Brand Names & Logos

Examples:

- Microsoft Windows → Copyright
- RSA Algorithm → Patent
- Apple Logo → Trademark

DMCA and DRM

Digital Millennium Copyright Act (DMCA)
Protects digital copyrighted content.

Prohibits:

- Illegal copying
- Circumventing protection systems

Exceptions:

- Fair Use
- Security Testing
- Encryption Research

Digital Rights Management (DRM)

Controls access and usage of digital content.

Examples:

- Netflix
- Spotify
- Kindle Books

SECTION 3

Privacy

Importance of Privacy

Privacy means protecting personal information.

Examples:

- Name
- Address
- Phone Number
- Health Records
- Financial Data

Why Important?

- Prevent identity theft
- Maintain trust
- Protect individual rights

Privacy Laws and Regulations



EU Data Protection Principles

- Notice
- Consent
- Access
- Security
- Enforcement

US Privacy Laws

- * Privacy Act 1974
- * HIPAA
- * COPPA
- * Electronic Communications Privacy Act

Organizational Response to Privacy

Organizations should:

- Develop privacy policies
- Appoint data protection officers
- Train employees
- Encrypt sensitive data
- Follow regulations

Privacy and Data Surveillance

Modern surveillance sources:

- Internet activity
- Mobile phones
- Social media
- Online payments
- Sensors & IoT devices

Privacy Protection Techniques:

- Encryption
- Anonymization
- Audit Logs
- Access Control

SECTION 4

Ethical Issues

What Are Ethics?

Ethics are moral principles that guide behavior.

Law tells us:

"What we must do"

Ethics tells us:

"What we should do"

Important distinction.

Ethical Issues in Information Systems

Common Issues:

- Unauthorized access
- Data misuse
- Privacy violations
- Software piracy
- Cyberbullying
- Manipulation of information

Discussion Question:

"Just because something is legal, is it always ethical?"



Professionals

Professionals should:

- Be honest
- Protect privacy
- Maintain confidentiality
- Avoid conflicts of interest
- Ensure public safety
- Use technology responsibly

Codes of Conduct

Professional Organizations:

- Association for Computing Machinery (ACM)
- Institute of Electrical and Electronics Engineers (IEEE)
- Association of Information Technology Professionals (AITP)

Common Principles:

- Integrity
- Honesty
- Fairness
- Responsibility
- Confidentiality

CASE STUDY

Scenario:

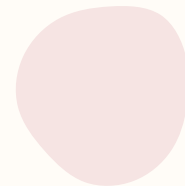
A software engineer discovers a security vulnerability before product release.

Options:

- * Stay silent to meet deadline
- * Report issue and delay release

Question:

What is the ethical decision?



Conclusion

Key Takeaways:

- Cybercrime threatens organizations and individuals
- Intellectual Property protects digital creations
- Privacy is a fundamental right
- Ethical behavior is essential for IT professionals
- Laws and ethics together create a secure digital environment